Eco-Loop

Autonomous Physical AI for Smart Building HVAC Optimisation

PROBLEM STATEMENT ID

No. 1 Type: MS

PS TITLE

Autonomous AI HVAC Control System

PS CATEGORY

Software

THEME

Smart Buildings & Energy Efficiency

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github.com/Arnavshukla09/Eco_Loop_Building_Agents

IDEA TITLE

Eco-Loop: Autonomous Physical AI for Smart HVAC Optimisation

🧠 Local LLM

Qwen2.5-7B via Ollama

🔌 MCP Server

10 Building Automation Tools

🏢 EnergyPlus

Real Physics Simulation

📊 Live Dashboard

React + WebSocket

One closed loop. Every 30 seconds. Sense → Analyse → Decide → Actuate. Fully autonomous, fully local.

🎯 **Estimated energy savings: 16.9% vs fixed-setpoint baseline | CO_2 avoided: ~930 kg/year per building**

PROPOSED SOLUTION

Describe your Idea / Solution / Prototype

Detailed Explanation

Eco-Loop is an autonomous Physical AI system that continuously controls HVAC setpoints in a multi-zone office building using a local open-source LLM.

The system operates in a 30-second closed loop:
1. EnergyPlus simulation provides real-time sensor readings
2. MCP server exposes 10 building tools to the AI agent
3. Qwen2.5-7B (via Ollama) analyses conditions + issues JSON decisions
4. Setpoints are applied to each zone independently
5. Results broadcast live to the React dashboard via WebSocket

How It Addresses the Problem

Buildings waste 30–40% of energy from fixed HVAC schedules that ignore occupancy, weather, and real-time conditions.

Eco-Loop solves this by:
Real-time adaptive setpoints per zone (not global fixed)
Occupancy-aware: widens deadband when building is empty
Weather-responsive: anticipates heating/cooling demand
Orientation-aware: perimeter zones treated independently
Predicted savings: 16.9% energy, 21.6% peak demand reduction

Innovation & Uniqueness

🔒 100% Local — No cloud APIs. LLM runs on-device via Ollama. Zero data privacy risk.
🔌 MCP Protocol — Building exposed as AI tools, compatible with ANY MCP client (Claude, Cursor, GPT).
🧪 Dual Mode — Seamlessly switches between real EnergyPlus and built-in physics simulator.
💬 Explainable — Every decision includes plain-English reasoning and confidence score in the dashboard.

TECHNICAL APPROACH

Technologies, Frameworks & Stack

Backend

- ✓ Python 3.11
- ✓ FastAPI
- ✓ asyncio
- ✓ Pydantic v2
- ✓ uvicorn

AI / LLM

- ✓ Qwen2.5-7B (Ollama)
- ✓ FastMCP 3.x
- ✓ Structured JSON output
- ✓ Rule-based fallback

Simulation

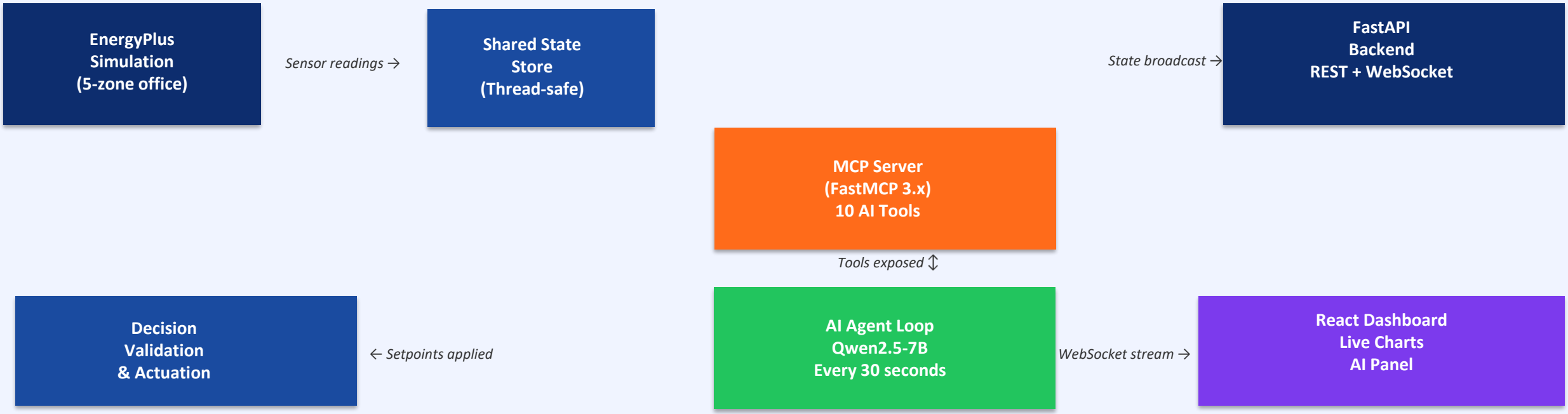
- ✓ EnergyPlus 24.1
- ✓ Python EnergyPlus API
- ✓ Custom Physics Engine
- ✓ Fanger PMV/PPD model

Frontend

- ✓ React 18 + Vite
- ✓ TailwindCSS
- ✓ Chart.js + react-chartjs-2
- ✓ WebSocket (real-time)

TECHNICAL APPROACH

System Architecture & Control Flow



Closed-Loop Cycle (every 30 seconds)

① EnergyPlus steps → temperatures, energy, comfort

② MCP tools called → structured building snapshot

③ Qwen2.5 LLM analyses → JSON setpoint decision

④ Validation → apply to all 5 zones

⑤ WebSocket → dashboard updates in real time

FEASIBILITY AND VIABILITY

✅ Feasibility Analysis

- Full working prototype built and tested
- Backend: 8/8 Python modules import cleanly
- Frontend: React builds successfully (4.37s)
- Physics sim: Verified 0.17°C/step drift
- Ollama integration: Tested with Qwen2.5-7B
- MCP server: All 10 tools functional
- Runs on a standard laptop (no GPU required)
- Simulation speedup: 60× (1 day in 6 sec)

⚠️ Challenges & Risks

- EnergyPlus API: Complex Python bindings
- LLM latency: 2–5s per decision cycle
- JSON parsing: LLM may produce invalid output
- Multi-zone coordination: Conflicting setpoints
- Thermal physics: Accurate calibration needed
- Windows compatibility: Path handling issues
- Production scale: Single-building PoC only

🛡️ Mitigation Strategies

- Physics simulator fallback (always available)
- Rule-based agent fallback (no LLM needed)
- Pydantic schema validation on all LLM output
- Setpoint bounds enforced (16–22°C heat, 20–30°C cool)
- 30× thermal mass multiplier for accuracy
- Cross-platform path handling in config
- Modular design: plug in real BMS sensors

ARTIFACTS

Source Code Architecture — 33 Files, ~160KB

File / Module	Purpose	Status
backend/energyplus_runner.py (23KB)	EnergyPlus API + physics sim dual engine	✅
backend/llm_agent.py (12KB)	Ollama/Qwen2.5 + rule-based fallback agent	✅
backend/mcp_server.py (12KB)	FastMCP server — 10 building automation tools	✅
backend/controller.py (9KB)	Autonomous control loop (sim + AI + actuation)	✅
backend/app.py (10KB)	FastAPI: REST + WebSocket + MCP mount + SPA serve	✅
backend/state.py (9KB)	Thread-safe singleton state store	✅
backend/dashboard_api.py (4KB)	WebSocket connection manager + broadcast loop	✅
frontend/src/App.jsx (7KB)	Main React app: header, grid, all components	✅
frontend/src/components/ (28KB)	7 components: MetricCard, AIPanel, Charts, ZoneGrid	✅
config/settings.py (6KB)	Pydantic-based settings (all env-var driven)	✅
energyplus/building.idf (12KB)	5-zone office EnergyPlus input file with EMS	✅
run.py (9KB)	Single-command launcher with dependency checks	✅

ARTIFACTS

Key Code Snippets

MCP Tool — read_building_state (mcp_server.py)

```
@mcp.tool()
async def read_building_state() -> dict:
    """Read full building state — all zones."""    obs =
state_store.get_observation()
    return {
        "zones": {z: {"temperature": s.temperature,
                      "setpoint_cooling": s.setpoint_cooling,
                      "pmv": s.pmv_score}
                  for z, s in obs.zones.items()},
        "hvac_power_kw": obs.hvac_power_kw,
        "total_energy_kwh": obs.total_energy_kwh,
        "outdoor_temp": obs.outdoor_temp,
        "occupancy_pct": obs.occupancy * 100,
    }
```

AI Agent Decision (llm_agent.py)

```
async def analyze_and_decide(obs, history):
    prompt = self._build_prompt(obs, history)
    response = await self._client.chat(
        model=settings.LLM_MODEL,
        messages=[{"role": "user",
                   "content": prompt}],
        format=AgentDecision.model_json_schema(),
        options={"temperature": 0.3},
    )
    return AgentDecision.model_validate_json(
        response.message.content
    )
```

Sample AI Decision JSON Output (Qwen2.5-7B response)

```
{
  "zone_setpoints": { "Core": {"cooling": 25.0, "heating": 19.0},
                      "Perimeter_N": {"cooling": 24.0, "heating": 20.0},
                      "Perimeter_S": {"cooling": 25.5, "heating": 19.5} },
  "fan_speed": "medium",   "energy_mode": "eco",   "confidence": 0.88,
  "reasoning": "Low occupancy (52%) detected at 14:00. Widening deadband on all zones to reduce compressor load.
               South-facing perimeter has solar gain — raising cooling SP 1.5°C. Est. savings: 14.2%.",
  "estimated_savings_pct": 14.2
}
```

ARTIFACTS

Dashboard Design & Simulation Metrics

🏢 **Eco-Loop Building Agent | Honeywell Automated Control | ● Live**

🌡️ Avg Temp	⚡ HVAC Power	💚 Comfort	💰 Savings	🌿 CO_2 Saved
22.3 °C	14.2 kW	87 %	16.9 %	8.2 kg

🤖 **AI Agent Decision | Mode: ECO | Confidence: 88%**

"Low afternoon occupancy at 52%. Widening thermal deadband on all zones.
South-facing perimeter raising cooling SP +1.5°C (solar gain detected).
Fan reduced to medium. Est. savings this cycle: 14.2%."

💚 Comfort Score: 87%
PMV: +0.15 (Neutral)
ASHRAE 55: ✓ Compliant
PPD: 6.2%

Total Energy (kWh)	Baseline (kWh)	Peak Demand (kW)	Baseline Peak	Energy Saving	CO_2 Avoided/yr
125.3	142.5	19.2	24.5	16.9%	930 kg

RESEARCH AND REFERENCES

📐 Standards & Comfort Models

ASHRAE Standard 55-2017: Thermal Environmental Conditions for Human Occupancy — Used for PMV/PPD comfort bounds and compliance thresholds.
ISO 7730:2005: Ergonomics of the thermal environment — Fanger PMV equation implementation.

🏢 Building Simulation

EnergyPlus v24.1 Documentation & Engineering Reference (US DOE, 2024) — Building physics, HVAC ideal loads, EMS actuator API.
https://energyplus.net/documentation

🤖 AI / LLM

Qwen2.5 Technical Report — Alibaba DAMO Academy (2024) — LLM used for decision-making.
Ollama Documentation — Local LLM serving: https://ollama.com
Model Context Protocol Specification — Anthropic MCP (2024): https://modelcontextprotocol.io

🔌 Frameworks & Libraries

FastMCP 3.x Documentation — https://gofastmcp.com
FastAPI — https://fastapi.tiangolo.com
React + Vite + TailwindCSS — https://vitejs.dev | https://tailwindcss.com
Chart.js — https://www.chartjs.org

📊 Energy Research

US DOE: 'Buildings Energy Data Book' — Building sector: 40% of US energy consumption.
ACEEE: 'Energy Efficiency in Buildings' — HVAC accounts for 40% of building energy use.
IEA: 'Buildings — Tracking Clean Energy Progress 2024' — Global building emissions baseline.

🔗 Project Repository

GitHub: https://github.com/Arnavshukla09/Eco_Loop_Building_Agents
All source code, documentation, IDF models, and demo scripts available open-source.